
Auditing Density Uncertainty in Cryo-EM Beyond a Gaussian Reconstruction Dictionary

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Abstract

A confidence interval can be calibrated within a reconstruction dictionary and still miss errors excluded by that dictionary. We audit fixed linear density features in cryogenic electron microscopy (cryo-EM) through the continuous Fourier-slice adjoint, separating a Gaussian fitting representation from the density class covered by inference. We specialize classical bias-aware intervals to supported L^2 density, derive bounded nonlinear-pose corrections with integrated Fourier-column moments, and implement matrix-free optimization with explicit integration remainders and primal-dual diagnostics. Two frozen studies use acquisition geometries from three experimental stacks, 144 audit settings and 3,456 signal/scenario records. The original pose audit has minimum in-class coverage 0.9938; a later moment refinement reduces no-data fallbacks from 48 to 16 on its separate 48-setting study. A classical testing lower bound places fixed-pose widths within a factor 1.20 of the optimal deterministic length in the declared class. These guarantees expose substantial limits: fine-feature detection can remain powerless, pose bounds are conservative, and experimental background dependence challenges whitening. Real-particle reconstructions and fresh-exposure prediction comparisons complement, but do not establish, density coverage. Density radii, support, noise and pose bounds remain explicit assumptions requiring experimental calibration.

¹Fourier Cryo Splats research project. Correspondence to: Project repository <<https://github.com/yashpatel5400/fourier-cryo-splats>>.

1. Introduction

Cryo-EM reconstructs molecular density from noisy projections whose orientations, translations and imaging parameters are imperfectly known. Modern neural and Gaussian representations offer expressive reconstruction models, but expressiveness alone does not establish which reconstructed features the data support. We focus on a fixed local density average or contrast. This target differs from a conformational population, the uncertainty of an atomic model, or prediction of a noisy held-out particle image.

An uncertainty calculation usually inherits its reconstruction space. If a Gaussian dictionary spans S , a posterior or confidence interval computed in S need not cover density outside S . Improving optimization inside the dictionary does not remove this discrepancy. Independent half reconstructions can also share the same representational restriction. Our working question is whether one can retain Gaussian computation while auditing feature uncertainty in a larger, explicitly declared density class.

We develop three connected components: a continuous Fourier adjoint-residual audit that separates fitting and inference spaces; bounds for local pose errors, including shared density–pose interactions and an integrated nonlinear remainder; and frozen validation protocols that distinguish fixed-parameter coverage, Bayesian prior-predictive coverage and experimental-image prediction. General bias-aware intervals, norm duality, moment inequalities and primal-dual optimization are established tools. The contribution is their explicit continuous Fourier-slice specialization, its computational implementation, and an empirical audit of when the resulting guarantees remain informative.

2. Related Work and Scientific Scope

Probabilistic reconstruction. Bayesian inference is foundational in cryo-EM (Scheres, 2012). Ullrich et al. (2020) already combine differentiable Fourier-slice reconstruction with variational density uncertainty and discuss the distinction between variance and model bias. Rangan et al. (2024) examine soft modes of the joint volume/alignment likelihood Hessian. Thus neither a probabilistic Fourier representation nor pose–density ambiguity is a new observation.

Our candidate emphasizes a finite-sample, bounded-class statement and a check outside the fitting dictionary.

Representations and heterogeneity. CryoDRGN learns a latent-conditioned density family (Zhong et al., 2021a); its latent variation is not automatically calibrated uncertainty in a homogeneous density. Ab initio neural inference (Zhong et al., 2021b) is a different task from fixed-pose backprojection. Gaussian mixtures and Gaussian splatting are also established (Chen & Ludtke, 2021; Chen & Ling, 2026; Qu et al., 2025; Shekarforoush et al., 2025). We use Fourier kernels as a numerical basis, without treating their existence as the contribution. Continuous-state recovery, conformational populations and pose entanglement are substantial separate problems; the living survey records current RECOVAR, DynaMight, cryoDRGN-AI and CAHRA work with source-level distinctions.

Validation and map confidence. Gold-standard splitting and noise substitution address reconstruction overfitting (Scheres & Chen, 2012; Chen et al., 2013); a shell correlation is not a coverage statement for a local feature. LocScale-2.0 provides calibrated confidence for map enhancement relative to its reference target (Bharadwaj et al., 2026). That target must not be conflated with confidence in an unknown density from raw particles. Map significance relative to background noise, directional resolution, atomic-model reliability and particle-derived confidence likewise answer different questions.

Statistical foundations. Optimal recovery links linear inverse estimation to bias–variance tradeoffs (Donoho, 1994). Honest inference over convex function classes and bias-aware Gaussian critical values are classical (Armstrong & Kolesár, 2018). Our theorems below specialize these principles and identify the assumptions that cryo-EM introduces. The survey and its bibliographic ledger are ongoing; a down-loaded paper is not counted as a completed critical review.

3. Forward Model and Inference Target

With $F(k) = \int \rho(x) e^{-2\pi i k^\top x} dx$, a particle obeys

$$Y_i(q) = C_i(q) e^{-2\pi i q^\top t_i} F(E_i q) + \epsilon_i(q). \quad (1)$$

E_i embeds a detector plane, C_i is its contrast transfer function (CTF), and frequencies and translations use reciprocal units. We realify one representative of each conjugate Fourier pair and whiten the noise. The initial theory conditions on a fixed design and an independent pilot ρ_0 , giving

$$y - \mathcal{A}\rho_0 = \mathcal{A}\delta + Ju + r + \epsilon, \quad \epsilon \sim N(0, I). \quad (2)$$

Here $\delta = \rho - \rho_0$, J is block diagonal over particles, u describes bounded nuisance errors, and r is a bounded re-

mainder. We assume

$$\|\delta\|_H \leq B, \quad \|u_i\| \leq \eta_i, \quad \|r_i\| \leq \gamma_i. \quad (3)$$

H is a declared Gaussian subspace, a supported voxel space, or continuous L^2 on a stated cube. These are different classes. A physical density-energy norm includes voxel volume; experiments using an ordinary voxel Euclidean norm explicitly normalize that norm. B , η_i and γ_i are assumptions, not quantities automatically learned from residual fit.

The estimand is $t = \langle \ell, \rho \rangle$, such as a Gaussian-weighted spatial average or the difference of two such averages. The smoothing scale and coordinate frame are specified in physical units. For weights fixed without the inference noise, use

$$\hat{t} = \langle \ell, \rho_0 \rangle + w^\top (y - \mathcal{A}\rho_0). \quad (4)$$

The pilot, dictionary, whitening, target and weights may depend on independent development data. Poses estimated from the inference images violate literal fixed-design independence; supplied consensus poses therefore support only a conditional experimental analysis here.

4. Bias-Aware Confidence and Ambient Auditing

4.1. A conditional certificate

Define $s = \|w\|$ and

$$b_H(w) = B \|\ell - \mathcal{A}^* w\|_H + \sum_i \eta_i \|J_i^\top w_i\| + \sum_i \gamma_i \|w_i\|. \quad (5)$$

Let $q(s, b, \alpha)$ solve

$$\Phi((q - b)/s) - \Phi((-q - b)/s) = 1 - \alpha \quad (6)$$

when $s > 0$, and set $q(0, b, \alpha) = b$.

Theorem 1 (Conditional coverage). *Under Equations (2)–(3), the interval $\hat{t} \pm q(s, b_H(w), \alpha)$ covers t with probability at least $1 - \alpha$, uniformly over the stated class, conditional on the independent design choices.*

Proof. The deterministic estimation error is $\langle \mathcal{A}^* w - \ell, \delta \rangle + \sum_i w_i^\top J_i u_i + \sum_i w_i^\top r_i$. Cauchy–Schwarz bounds its absolute value by b_H . The remaining error is Gaussian with standard deviation s . Coverage of a symmetric interval decreases with the absolute Gaussian mean, so the minimum occurs at mean $\pm b_H$ and equals Equation (6). $\square$

The result is a direct bias-aware inference construction. It is useful only if its assumptions are defensible and its width improves on the no-data interval $\langle \ell, \rho_0 \rangle \pm B \|\ell\|$.

4.2. Why auditing the dictionary matters

Let $S : \mathbb{R}^p \rightarrow H$ have orthonormal columns. A restricted calculation replaces the density term by $B \|S^*(\ell - \mathcal{A}^*w)\|$. Write $h = \ell - \mathcal{A}^*w$. Then

$$\|h\|_H^2 = \|S^*h\|^2 + \|(I - SS^*)h\|_H^2. \quad (7)$$

The second term measures uncertainty discarded by the dictionary. It can be large even when the restricted residual is zero. Evaluating the full adjoint residual repairs this omission for H without estimating a representation-error direction from noisy data. It does not cover density outside the declared ambient grid or support.

Centering in Equation (4) also uses the ambient operator. If a surrogate centers the images instead, the discrepancy $w^\top(\mathcal{A} - \bar{\mathcal{A}})\rho_0$ requires an additional bound. We do not silently substitute a surrogate forward model into the coverage theorem.

4.3. Optimization and verifiable stopping

We optimize the conservative convex objective

$$\mathcal{F}_H(w) = z_{1-\alpha/2} \|w\| + b_H(w), \quad (8)$$

then report the sharper width from Equation (6). Its dual is

$$\begin{aligned} \max_{v,a,d} \quad & \langle \ell, v \rangle \\ \text{s.t.} \quad & \|v\| \leq B, \quad \|a_i\| \leq \eta_i, \quad \|d_i\| \leq \gamma_i, \\ & \|\mathcal{A}v - Ja - d\| \leq z_{1-\alpha/2}. \end{aligned} \quad (9)$$

Representing each norm by a Euclidean support function proves this dual; continuity and coercivity yield strong duality. A feasible dual iterate supplies a lower bound before convergence. The gap concerns Equation (8), not optimality of the folded-normal width.

For a reduced density space S , a feasible dual vector lifts to Sv in H . Thus a restricted dual lower bound and an ambient primal upper bound bracket the same full-space optimum. We enrich S with the component of $\ell - \mathcal{A}^*w$ orthogonal to it, using only design and target information. Restricted exact optima increase under enrichment; individual ambient primal values need not decrease, so we keep the best feasible iterate. No convergence-rate claim is made for this greedy construction.

Quadratic majorization gives coefficient-space conjugate-gradient solves with particle-block inverses. A classical Chambolle–Pock implementation is an alternative (Chambolle & Pock, 2011). For fixed poses, a full-space matrix-free comparator solves $(\mathcal{A}^*\mathcal{A} + \lambda I)v = \ell$ within majorization. It can outperform enrichment and is included to prevent an unsupported Gaussian speed claim.

5. Continuous Density and Bounded Pose Errors

5.1. A Fourier audit independent of the fitting dictionary

For fixed poses, the audit also extends to $H = L^2([-1/2, 1/2]^3)$ without a voxel or density bandwidth assumption. This cube is a broader support class than the earlier spherical mask. Its Fourier kernel is $K(v) = \prod_a \text{sinc}(v_a)$. If $c_j = (w_{Rj} + iw_{Ij})C_j/\sigma$ and $f_w(x) = \text{Re} \sum_j c_j e^{2\pi i k_j^\top x}$, then

$$\|f_w\|_H^2 = \frac{1}{2} \text{Re} \sum_{j,l} \{c_j \bar{c}_l K(k_j - k_l) + c_j c_l K(k_j + k_l)\}. \quad (10)$$

Gaussian target norms and cross terms have analytic truncated-normal integrals, giving $\|\ell - \mathcal{A}^*w\|_H$ directly. The continuous objective has a feasible dual vector $v = th$, where $h = \ell - \mathcal{A}^*w$ and $t = \min(B/\|h\|, z/\|\mathcal{A}h\|)$. Its lower bound is $t(\|\ell\|^2 - w^\top \mathcal{A}\ell)$.

A matrix-free variant integrates only the finite Fourier adjoint field with Gauss–Legendre quadrature. An explicit quadrature remainder protects both primal and dual bounds; it does not discretize the unknown density class. Standard pivoted Cholesky preconditioning accelerates the solves (Harbrecht et al., 2012). The formulas and error derivation are in Appendix E. A separate continuous nonlinear-pose extension bounds the norm of the first and second polynomial Fourier fields by padded quadrature Grams. Each particle has a normalized five-dimensional pose vector u_i in the unit ball: three rotation coordinates with radius a and two detector shifts with radius s_t . For fixed w , first and second adjoint derivatives form five linear and fifteen symmetric quadratic Fourier fields per particle. Their combination at allowed u_i is bounded by a shared-field norm L_{12} , using spectral and triangle bounds with integration padding. The unknown density is shared across particles; treating it separately in each image would lose that structure. Finite Gaussian/voxel derivations and alternatives appear in Appendix A. With $P = \|\rho_0\|$ and their shared-field bound L_{12} , its bias bound is

$$b \leq B \|h\| + (B + P) \left(L_{12} + \frac{1}{6} \sum_i K_{3i} \|w_i\| \right). \quad (11)$$

Here K_{3i} uniformly bounds the continuous third-derivative operator. Appendix E.1 derives the constants and integration error. The current implementation post-audits fixed-pose weights and uses a conservative pilot norm; no nonlinear optimality claim follows.

An integrated-moment refinement replaces the spatial-

maximum cubic remainder by

$$r_{\text{mom}}(w) = \frac{B+P}{6} \sum_{i,q} |c_{iq}| (L_6^3 + 3L_4H_4 + T_2). \quad (12)$$

Here $L_p = 2\pi(a\|k\|c_p + s_t\|q\|)$, $H_p = 2\pi a^2\|k\|c_p$, $T_2 = 2\pi a^3\|k\|c_2$, and $(c_2, c_4, c_6) = (12^{-1/2}, (3/12^2)^{1/4}, (15/12^3)^{1/6})$. These constants bound the cube-integrated moments of linear spatial coordinates. Hölder’s inequality controls the exact third phase derivative; retaining actual frequency weights avoids a Hilbert–Schmidt relaxation (Appendix E.2). No distribution on the unknown density is introduced.

Operationally, we fix the pilot, target, noise model and density/pose classes independently of inference noise; optimize fixed-pose weights; audit their continuous residual and pose derivatives; then use the smallest of the valid remainder bounds. If the resulting interval is wider than the no-data alternative, we switch both center and width to that alternative using design information alone. Only then are inference images used in Equation (4). The moment refinement is evaluated in a separate frozen follow-up, preserving the original study.

6. Experiments

Results below distinguish exploratory development from three completed protocols frozen before their respective outcomes: original continuous uncertainty validation, an integrated-moment follow-up, and additional-exposure prediction. Experimental development pools had already been inspected; only the specified frozen follow-ups constitute prospective tests.

6.1. Data, references and provenance

We retain 8,192 extracted experimental images from each of EMPIAR-10028, 10049 and 10076 (EMPIAR, 2015a,b; 2016). Source exposure groups are reconstructed from deposited metadata. EMPIAR-10076 requires the original Frealign FILM field because its STAR micrograph numbers are sequential particle identifiers. Hash-based whole-group partitions provide pilot, tuning, inference-half and test labels within this development pool. Consensus poses and CTFs remain conditional inputs.

Deposited maps EMD-2660, EMD-6487 and EMD-8434 define independent voxel signal generators after explicit low-pass resampling. They are simulation truth, not proof of experimental accuracy. Nonuniform Fourier transforms project these maps independently of the Gaussian representation. Native and downloaded-file hashes, source indices, signs, field sizes and software versions are recorded.

The physical fields are 482.4, 236.16 and 419.2 Å, respectively. Gaussian feature standard deviations of 0.03/0.07

times the field therefore span 7.1–33.8 Å. The radius-five inference band reaches wavelengths 47.2–96.5 Å; the radius-twelve extension reaches 19.7–40.2 Å. These uncertainty experiments probe coarse structural features, not near-atomic reconstruction. The separate image-prediction comparison uses a wider common band on the 64×64 working grid.

6.2. Matched controls and exploratory finite-space studies

The numerical controls use identical density functionals and declared units: full/diagonal Gaussian posterior covariance, regularized sampling covariance, joint Gaussian pose covariance, source-group bootstrap, fixed-pose bias-aware intervals and the no-data estimator. Posterior methods pass the matched prior-predictive identity; fixed-parameter boundary coverage is a different criterion. These are mathematical controls, not claims to reproduce every external method’s complete implementation.

The exploratory studies in Appendix B motivate the continuous extension. A dictionary can exclude measurable bias even when an interval is calibrated in that dictionary. Quadratic pose terms and shared-density spectral bounds improve a first-order relaxation. Weakly regularized full-voxel bootstrap coverage is 0.938/0.956/0.958 with smaller widths than the robust audit; stronger regularization can reduce coverage to 0.338. Thus the audit does not uniformly dominate empirical interval procedures in width. Grid agreement alone does not prove a continuous-density guarantee.

6.3. Frozen continuous-density validation

The first protocol was frozen at commit 76dbd53 before its outcomes. Four additional-cohort geometry subsets per stack, two feature scales, two targets and two pose budgets give 96 audit settings and 2,304 signal/scenario records. Each subset contains 128 particles; its prescribed density radius is $B = 2$, and the simulated Gaussian noise scale is fixed from a pilot SNR convention. Four unit-norm cell generators include a deposited map, ellipsoid unions, modulated shells and signed multiscale fields. Exact continuous-density boundaries exercise directions outside the cell representation. The protocol includes nominal, random-boundary and coherent-boundary poses, plus separately reported four-fold pose exceedance and doubled noise scale. Acquisition subsets may share exposure groups; they are not independent biological replicates.

All 48 unique fixed-pose fits meet the 0.5% sum-objective gap. Across all 1,536 in-class signal/scenario records, minimum analytic coverage is 0.99377 for the continuous pose audit, 0.93922 for the fixed-pose audit and zero for the 24-cell projection audit. The cell-defined generators are covered conservatively by both audited fixed-pose procedures; the severe cell-audit failures occur at explicitly continuous

Table 1. Frozen continuous-density and pose validation: relative half-width ranges over both targets and all four geometry replicates per stack. All 48 settings and all in-class signals are retained. The density, noise and pose limits are supplied assumptions.

EMPIAR	Scale 0.07	
	1°	2°
10028	0.39–0.50	1.00–1.00
10049	0.38–0.47	1.00–1.00
10076	0.30–0.36	0.87–0.99

boundaries. These are worst-case checks, not typical-map failure estimates. Each case also uses 10,000 paired scalar Gaussian draws to check analytic noise-marginal coverage.

Figure 1 shows all selected designs. Broad-feature relative widths are 0.109–0.175 at 0.1° and 0.271–0.457 at 0.5° . Fine-feature widths rise to 0.759–0.779 and 0.885–0.998, respectively, nearly reaching the no-data alternative. None of the 96 original-audit settings switches to no data. The 768 deliberate model-violation records also remain conservatively covered by the pose audit (minimum 0.99948); these controls do not expose failure at the tested severities. They do not validate the violated assumptions. Sign power, marginal Monte Carlo intervals, all methods and every failure remain in the released records.

6.4. Frozen integrated-moment follow-up

The follow-up was frozen at 8d66785 after the refinement’s development, with fresh signal/noise seeds and particle subsets disjoint from the original study. Its 48 settings use broad targets and $1^\circ/2^\circ$ budgets. All 24 unique fits converge within the 0.5% gap. Across 768 in-class records, the refined audit’s analytic coverage rounds to one in float64, whereas the fixed-pose audit reaches 0.67114 and the cell-projected audit 0.12596. The 384 model-violation records also remain covered by the refined audit at the tested severities.

The spatial-maximum bound uses no data in all 48 settings; integrated moments reduce this to 16 (Table 1). All 1° widths are 0.304–0.501 of no data. At 2° , only EMPIAR-10076 remains narrower, at 0.866–0.987. However, correct-sign power for the in-class cell generators is at most 2.3×10^{-5} ; constructed continuous boundaries attain at most 0.5184. Thus the gain in interval width is real but does not demonstrate useful detection in these map-like signals. Figure 3 and released records retain all settings.

6.5. Continuous audit and assumption failures

The continuous fixed-pose comparison uses the same three acquisition geometries, two targets and two averaging scales. Pilots and independent map generators are orthonormal constant-cell functions on grids 24 and 64, respectively.

Their exact Fourier transforms include cell sinc factors and a half-cell phase; target integrals use normal-CDF differences. Both have unit continuous L^2 norm, so the same prescribed $B = 2$ covers their difference. The inference class itself contains all signed L^2 functions on the cube; no positivity or atomic-structure restriction is imposed.

All 12 dense continuous solves meet the 0.5% sum-objective gap, and independently evaluated matrix-free results agree to numerical tolerance. Figure 7 distinguishes the continuous bound from projections onto finite cell spaces. At a constructed continuous bias boundary, small relative width differences can matter: a 64-cell interval at the EMPIAR-10028 broad central estimator has coverage approximately 0.923 despite a width difference of only 1.7%. The independent map’s coverage remains one; this adversary is not a typical-signal claim. Refitting weights for the larger cube class is essential: transferring sphere-only weights can be much less informative.

A classical Gaussian two-point lower bound supplies an additional design-only efficiency check (Corollary 1). Across all 18 low- and higher-band fixed-pose targets, its necessary half-length is 0.840–0.882 of the reported interval half-width. Thus the fixed-pose procedures are within a factor 1.20 of the minimum possible deterministic interval length under the stated continuous class, subject to the reported numerical accuracy. This compares against arbitrary interval centers, but not variable-length intervals. The factor is consistent with the classical testing/duality connection below; it is not a new minimax principle. It does not imply that the additional pose bound is comparably sharp.

The first 128-particle timing comparison uses about 419 MB for dense Gram arrays, versus 86 MB for quadrature plus a 1,024-column preconditioner. Setup plus solve/audit takes approximately 14 versus 6 seconds under concurrent Mac load; these stored-array counts are not peak resident memory. An initial weak preconditioner was substantially slower and is retained. The matrix-free implementation also completed both broad targets with 1,024 particles for all three geometries. A separate radius-12 continuous comparison completes both fine targets on all three geometries: relative widths range from 0.120 to 0.238, with all six sum-objective gaps below 0.5%. Individual solve/audit times range from 483 to 1,223 seconds. Despite improvement over no data, reference-generator sign power is below 1.1×10^{-9} in every fine-target case. These results do not establish biologically useful fine-feature detection. Projecting the same residual onto 64-cell densities yields coverage 0.791–0.894 at the constructed continuous boundary, despite widths only 3.7–6.5% smaller.

The joint continuous-pose post-audit is complete for both broad targets, all three geometries and rotation budgets $0.1^\circ/0.5^\circ$, with a joint shift budget of 0.01/24 field units.

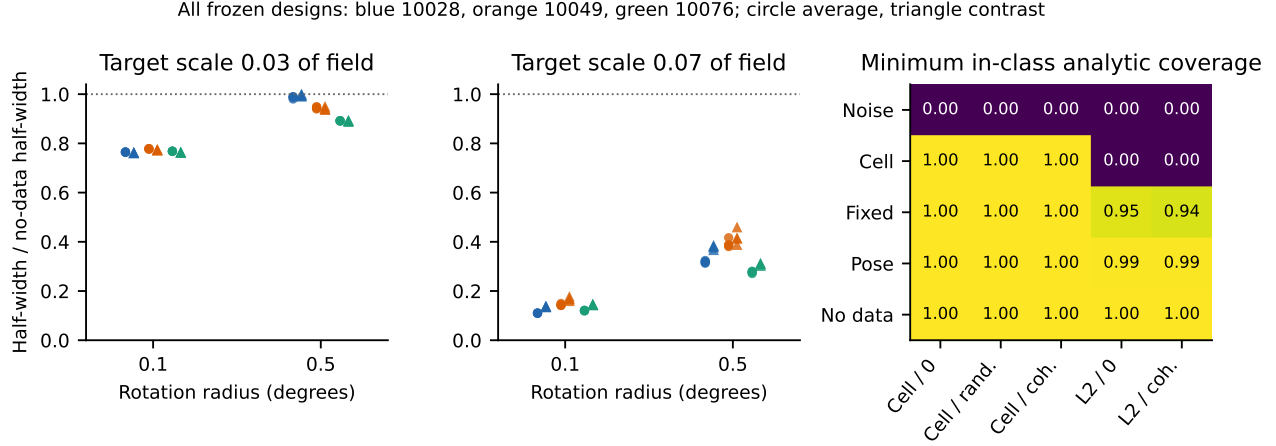


Figure 1. Completed frozen original-audit study: all 96 settings. Each geometry contributes both feature targets at each scale and pose budget. Width is relative to the same density class’s no-data interval. The right panel reports minimum analytic coverage by generator/scenario group; “L2” denotes constructed continuous-density boundaries. Values are rounded for display. Fine-feature widths near one and conservative cell-generator coverage limit a usefulness interpretation.

Its twelve relative half-widths range from 0.110 to 0.444 (Figure 8). Exact nonlinear constant-cell projections give reference coverage at least 0.99999999998 in these tests, revealing conservatism rather than empirical calibration. At one nonzero coherent pose per case, analytic sinc integrals eliminate the entire continuous density ball; its feasible bias is 0.24–0.86 of the uniform upper bound. These are lower bounds on worst bias, not globally optimized adversaries. Each audit takes about 38–40 seconds and stores approximately 725 MB of field/Gram arrays under concurrent Mac load.

Six explicit assumption-stress runs each retain 273 records (Figure 9). Support-only violations preserve the full density-energy radius, isolating the damage caused by an incorrect mask. View-coupled two-state mixtures preserve an equal-population mean but violate homogeneity. Both can cause severe undercoverage, as can correlated noise and understated noise scale. The tested low-bandwidth defocus offsets and elliptical Student- t controls do not cause undercoverage for the shared-density pose bounds. Every severity setting and unfavorable result is retained.

A wider development sweep exposes the original continuous bound’s limits: every broad target falls back to no data at 1° , 2° and 5° . The integrated-moment refinement reduces cubic remainder bounds by 6.79–14.74 times across all 30 small- and large-angle settings, without changing weights. No-data fallbacks decrease from 18 to 10; all six 1° intervals have relative widths 0.300–0.532 (Figure 4). This is post-hoc development, not the frozen study’s outcome. The remaining gap to feasible nonlinear biases and the large-angle failures are retained.

A stronger continuous stress search uses three starts for each

bias sign, both targets, all three geometries and $0.5^\circ/1^\circ/2^\circ$ budgets. All 108 final candidates are recomputed with analytic continuous-density integrals and exact nonlinear poses in float64 after an accelerated quadrature-surrogate search. The strongest feasible bias in each setting is 1.06–2.54 times its coherent-pose control, but only 0.16–0.57 of the integrated upper bound (Figure 5). None exceeds that bound. These local searches expose remaining slack; they neither prove a global maximum nor supply validated floating-point lower bounds.

6.6. Experimental reconstruction controls

We execute stock cryoDRGN 4.3.1 homogeneous neural training on both exposure-group halves of all three stacks: three hidden layers, width 256, batch size eight, fixed consensus poses and CTFs. These are actual neural fits; they are not ab initio inference or uncertainty estimates. Gaussian and voxel ridge fits use the same halves. All methods are evaluated on identical held-out particles and 1,410 Fourier pairs per image inside radius 30 of the 64-pixel grid, with the same window and image-background normalization. Neural training itself uses the stock radius-32 loss and per-half global scaling, so training objectives are not identical.

Table 2 selects checkpoints using tuning-group error only, from runs through 60 epochs for EMPIAR-10028/10049 and 100 epochs for EMPIAR-10076. The selected epochs are 60, 50 and 100. The development plateau check motivated these budgets; it is not a preregistered stopping rule or a proof of global convergence. Exposure-group bootstrap comparisons, checkpoint hashes and direct-network versus map-Fourier checks are recorded with the results. FSC assesses conditional map agreement; neither it nor image

Table 2. Experimental-image prediction error on development test groups (normalized MSE; lower is better). Neural checkpoints minimize tuning-group error among completed runs: epochs 60, 50 and 100 respectively. All half-map FSCs remain above 0.143 through the sampled limit in the last column, which is not a measured crossing.

EMPIAR	Gaussian	Voxel	Neural	Limit (Å)
10028	0.8355	0.8417	0.8293	16.08
10049	0.9308	0.9316	0.9302	7.87
10076	0.8764	0.8806	0.8707	13.97

prediction labels true experimental density coverage.

A frozen follow-up evaluates these unchanged models on 4,096 additional particles per dataset from exposure groups absent from development (Table 5). Neural prediction has lower NMSE than both controls in all three datasets; all six prespecified paired group-bootstrap differences remain positive under the stated multiplicity adjustment. This confirms conditional prediction ranking on new exposures, not experimental density coverage or independent-pose FSC.

Post-hoc background diagnostics on all 12,288 new images show mean nearest-neighbor normalized pair products of 0.197–0.292 across the three stacks and two image axes (Figure 6). White-Gaussian controls with the same background centering and normalization are near zero. The annular background can contain molecular signal, ice and preprocessing artifacts, so this is not identification of the measurement-noise law. It does show why simple spatial scaling cannot be assumed to establish whitening; our supplied-noise simulations do not resolve that experimental issue.

6.7. Baselines and numerical checks

Implemented matched mathematical baselines include full Gaussian posterior covariance, diagonal variational covariance, regularized sampling covariance, joint Gaussian pose covariance, identifiable unregularized estimation, fixed-pose bias-aware intervals, coarse and structured nuisance bounds, and the no-data interval. These are not executions of every external paper’s original code. Full Gaussian posteriors pass their exactly matched linear prior-predictive identity at 0.95. Narrow priors can nevertheless fail bounded fixed-signal tests; wider priors and unregularized estimation can produce very wide intervals.

Independent conic optimization checks the majorization, primal-dual and ambient solvers on small instances. Derivative finite differences, nonlinear remainder tests, direct Fourier sums, adjoint identities and independent Gaussian-integral checks validate implementation components. The current suite has 48 passing tests, including a three-dimensional Fourier-transform check for externally gener-

ated maps. Tight FINUFFT tolerances are numerical checks, not interval-arithmetic guarantees on roundoff.

7. Limitations

The current evidence does not establish an end-to-end uncertainty method with estimated poses, noise law and density class. Local pose radii and the total density bound are assumptions. Selecting them from the same inference residuals would require new analysis; a narrow residual does not rule out an almost invisible density error. Learned support can exclude true structure. Gauge and hand ambiguity must be fixed or included in the target class.

Our three stacks supply distinct experimental geometries, not ground-truth density coverage labels. The uncertainty simulations use coarse features and prescribed noise; the total density ball includes signed functions and is deliberately broader than molecular structure. Its fixed-pose near-optimality result therefore measures performance in that class, not optimality under positivity or atomic constraints. Pose corrections remain loose, and low sign power limits structural use even when widths improve over no data. Source-group splits do not undo full-data consensus-pose dependence. Full external probabilistic reconstruction pipelines, joint alignment resampling and converged ab initio comparisons remain unexecuted; mathematical covariance controls are narrower comparisons. The supplied class assumptions and empirical background correlations prevent an end-to-end calibration claim. Simultaneous maps require multiplicity control, and the targeted literature survey is not exhaustive across all cryo-EM applications.

Reproducibility and Impact

The active research branch is `uncertainty-study` at <https://github.com/yashpatel5400/fourier-cryo-splats>. It preserves failed development approaches and distinguishes them from completed outputs. The earlier reconstruction release v0.1.0 remains available with its conditional FSC results; it is not the uncertainty study’s accuracy claim. No new conference submission or acceptance is implied by this format. Background PDFs are retained locally with source hashes rather than redistributed. Prematurely calibrated-looking maps can mislead structural interpretation; explicit assumptions and negative controls are central outputs of this work.

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A. Finite Representations and Local Pose Bounds

We use smooth conjugate pairs

$$\begin{aligned} F_c(k) &= \sum_j c_j g_j(k) + \bar{c}_j g_j(-k), \\ g_j(k) &= e^{-\|k - \mu_j\|^2 / (2\sigma_j^2)}. \end{aligned} \quad (13)$$

This enforces Hermitian symmetry. Hard cutoff kernels from the earlier reconstruction prototype are unsuitable for an unqualified Taylor bound; the uncertainty implementation uses full Gaussians. Their analytic whole-space Gram matrix provides density-energy coordinates $\delta = T\beta$ with $T^*GT = I$, so $\|\beta\| \leq B$ is a density-energy assumption rather than an arbitrary coefficient scale.

Parameterize local rotations by a Lie-algebra vector and translations by detector shifts. Normalize their joint vector so $\|u_i\| \leq 1$, with physical radii incorporated in the operator. With $D_{ia} = \partial A_i / \partial u_{ia}$ at the pilot pose,

$$\begin{aligned} A_i(u_i)(c_0 + \delta) &= A_i(0)(c_0 + \delta) + J_i u_i \\ &\quad + \sum_a u_{ia} D_{ia} \delta + e_i. \end{aligned} \quad (14)$$

If H_i uniformly bounds the pilot's second derivative along all allowed segments and K_i bounds the corresponding operator derivative, then $\|e_i\| \leq (H_i + BK_i)/2$. Define $T_i(w_i) = [D_{i1}^\top w_i, \dots, D_{iq}^\top w_i]$. The interaction contributes at most

$$B \sum_i \|T_i(w_i)\|_{\text{op}} \leq B \sum_i \|T_i(w_i)\|_F. \quad (15)$$

The implemented Frobenius relaxation is a sum of Euclidean group norms. It ignores some sharing of δ across particles and can be conservative. Charging the entire first-order interaction to an isotropic remainder was substantially worse in development.

The same construction operates in the full supported voxel space. For voxel position x , box length d , rotation radius a and shift radius s_t , set

$$L(q, x) = \frac{2\pi}{d} (a \|k\| \|x\| + s_t \|q\|), \quad (16)$$

$$H(q, x) = \frac{2\pi}{d} a^2 \|k\| \|x\|. \quad (17)$$

The absolute second derivative of its complex Fourier column is bounded uniformly by $|C_i(q)|[L(q, x)^2 + H(q, x)]$ before whitening. Summing this column bound against $|\rho_0(x)|$ gives the pilot bound; its frequency/voxel Frobenius norm gives the operator bound. Analytic first derivatives supply the full-density interaction. Thus the implemented joint certificate covers the stated ambient density ball and local pose balls simultaneously. Fixed CTFs and known whitened Gaussian noise remain assumptions; CTF error, window correlations and heterogeneous densities are not covered by assertion.

A.1. Curvature and shared-density post-audits

Retaining the quadratic pose terms replaces the second-order remainder with a uniform cubic one (Appendix C.4). It adds structured pilot and density Hessian terms, and can substantially reduce width. The density component can be tightened further: it is one shared error across particles, not one independent error per image. Let $h = \mathcal{A}^*w - \ell$ and collect blocks $H_0 = h$, $H_i = T_i(w)$ and $H_{n+i} = U_i(w)/2$, where U_i contains all ordered second-derivative adjoint columns. Their coefficients are 1, u_i , $u_i \otimes u_i$, each of norm at most one. For any positive fixed block scales d_j ,

$$\left\| \sum_j H_j v_j \right\| \leq \sqrt{\sum_j d_j} \|[H_j / \sqrt{d_j}]_j\|_{\text{op}}. \quad (18)$$

Multiplication by B bounds the combined density residual and interactions. The pilot and cubic remainder terms remain unchanged. Equation (18) is an elementary operator-norm inequality. We compare it with separate spectral and triangle bounds, use $d_j = \|H_j\|_F$ for nonzero blocks, and retain the smallest valid bound. This post-audit uses fixed weights and preserves coverage; it does not inherit an optimality certificate for the new width.

First and second derivative adjoints require only five first and fifteen distinct second Fourier moment fields per particle. Direct summation or batched nonuniform transforms evaluate these fields on a larger supported grid without its pixel-space derivative Grams. Physical voxel-volume scaling, the inference center and the target all use that audit grid. This verifies larger finite density spaces, not arbitrary unresolved continuous density.

B. Finite-Space Development Experiments

B.1. Representation and exact-dictionary controls

A noiseless projection audit with 256 particles per split and a radius-six frequency band found relative held-out errors of approximately 1.8%, 2.7% and 7.7% for the finest tested Gaussian dictionary. These residuals demonstrate why noise-only intervals do not cover representation error.

The bounded-model coverage benchmark has 48 settings: 12 seeds, broad versus preferred orientations, and rotation radii 0.5° or 2° . Each setting uses 96 particles, 40 coefficients, 20 random bounded signals and estimator-specific adversarial controls. We report exact Gaussian noise-marginal coverage and cross-check with 2,000 paired noise draws per signal. Shared designs and paired noise do not constitute independent voxel trials.

The exact-dictionary comparison is plotted in Appendix G. The structured certificate's smallest tested coverage was 0.991 at nominal 0.95; omitting the quadratic remainder

gave 0.932. Median width relative to the no-data bound was 0.557 for broad views at 0.5° , and approximately one for preferred views. This is a usefulness limitation, not a calibration success to conceal with pooled averages.

B.2. Ambient-space audit

The fixed-pose audit uses 128 real acquisition geometries per dataset, a 24^3 voxel grid, a spherical support of radius 0.35 times the field, and prescribed signal-to-noise power ratio 0.1. A pilot derived from real particles and each reference generator are normalized to unit voxel Euclidean norm. We set $B = 2$ before seeing coverage; the triangle inequality covers all such normalized generator/pilot pairs. This simulation convention does not estimate the density radius of experimental data.

The initial dictionary has 33 non-null Gaussian modes. Two targets, a central average and an axial contrast, use Gaussian averaging widths 0.03 and 0.07 times the field. Restricted intervals, the same weights with a full-grid audit, enriched weights, and the ambient matrix-free solution use identical estimands. The larger-grid model is deliberately able to express errors outside the initial dictionary.

Table 3 reports median widths across the two targets. The full-space solver reached its 0.5% objective-gap tolerance in 12 of 12 cases, taking 2.4–11.1 seconds per target on the Mac. The enrichment solver reached the same outer tolerance in 8 of 12 cases within 100 rounds; nonconverged results retain valid bounds but do not establish optimality. The solvers optimize the conservative sum-width, so their sharper reported widths need not be ordered exactly by their objective gaps.

Dictionary-only intervals can be narrow because they exclude admissible density errors. For the EMPIAR-10028 central broad average, analytic coverage on the independent voxel-map generator is approximately 4.2×10^{-5} . The same image weights, with the ambient bias bound, cover at the prescribed level over the ambient ball. On each estimator’s constructed ambient bias boundary, the audited intervals attain nominal 0.95 Gaussian coverage up to numerical tolerance. These boundary calculations check the theorem and implementation; they are not independent evidence of practical class calibration. Broader targets have substantially smaller relative widths than fine targets. The prescribed norm radius and fixed-pose assumption are essential to interpreting that result.

B.3. Joint ambient and nonlinear pose uncertainty

We combine the full supported voxel space with the nonlinear pose bound, using the broad axial contrast, 128 particles per geometry and the same prescribed density radius. Rotation radii are 0.1° , 0.5° and 2° , with a shift radius of 0.01

working pixels in their joint unit ball. Signal cases include random density/pose boundaries, estimator-directed adversaries and the independent voxel reference under a coherent rotation. Table 4 shows that a valid but conservative interval can still be uninformative at a large nuisance radius. Fixed-pose intervals reach coverage as low as 0.904 on these allowed perturbed cases. The nonlinear certificate’s smallest tested analytic coverage is 0.9997; its conservatism requires scrutiny rather than being interpreted as superior calibration. The earlier first-order optimization-budget comparison is retained in the appendix.

The completed curvature comparison spans three geometries, two targets and three pose radii. All 18 quadratic solves reach the 0.5% objective-gap tolerance. At 0.5° , their width/no-data ratios are 0.128–0.178, versus 0.423–0.532 at 2° . Figure 2 shows the full comparison, including probability of establishing the correct target sign. Central averages become detectable at the intermediate radius; the deposited-map axial contrasts remain too small. Their averaging standard deviation is 0.07 times the field (16.5–33.8 Å), so this does not establish atomic-scale utility.

For both targets at 0.5° and 2° , the shared-density post-audit reduces the same weights’ interval widths to 0.50–0.65 of their original values. Independent local nonlinear stress tests optimize pose errors while eliminating the density ball exactly. Four starts for each bias sign find feasible adversaries, recomputed in float64 after MPS search; they supply lower bounds on worst bias, not global solutions. Their comparisons with the upper bounds are retained in the repository.

A finer-feature development probe uses a 32^3 grid, frequency radius 12 and averaging width 0.03 times the field. At 0.5° on EMPIAR-10028, the first-order bound equals the no-data interval; retaining curvature reduces its relative width to 0.449. The completed comparison spans both targets and all three geometries: all twelve quadratic solves converge, with relative widths 0.166–0.231 at 0.1° and 0.442–0.477 at 0.5° . These remain finite-grid results.

B.4. Bootstrap tradeoff and finite-grid stability

A fixed-pose benchmark fits 33- and 123-mode Gaussian models and all 2,517 supported voxels. It uses 999 source-group bootstrap resamples, four prescribed ridge strengths and 1,000 independent Gaussian measurement-noise replications per setting. Percentile/basic intervals, sampling covariance, posterior covariance and a same-estimator ambient bias audit are recorded. The group-resampling plan is reused across noise draws; binomial error bars are conditional on that plan.

Weakly regularized full-voxel bootstrap intervals have center coverage 0.938, 0.956 and 0.958 across the three geometries,

Table 3. Fixed-pose ambient audit: median interval width relative to the no-data width across two density targets. All three methods after auditing use the same supported voxel class. The first column instead assumes the initial 33-dimensional dictionary is correct and is not an honest interval for that larger class. All entries are development results.

Dataset	Averaging width / field	Dictionary only	Same weights, audited	Enriched	Full matrix-free
10028	0.03	0.008	0.978	0.689	0.687
10028	0.07	0.007	0.510	0.053	0.050
10049	0.03	0.020	1.016	0.718	0.716
10049	0.07	0.020	0.636	0.075	0.073
10076	0.03	0.024	0.949	0.721	0.719
10076	0.07	0.023	0.462	0.072	0.072

with widths smaller than the robust audit. Strong regularization can cause substantial undercoverage: EMPIAR-10028 center coverage drops to approximately 0.338. A conservative bias audit therefore does not dominate bootstrap width in every setting. Figure 11 retains the regularization sweep, including cases favorable to bootstrap.

A fixed-pose refinement study audits 48 coarse-grid solutions on a common 64^3 supported space. Grids 16, 24, 32 and 48 preserve physical field, density-energy norm and noise units. Width ratios between fine audit and original calculation range from 0.72–1.45 at grid 16 and approximately 0.999–1.006 at grid 32. Nonlinear Fourier-moment audits additionally cover both targets, both pose radii and all three geometries on grids 24, 32 and 64. Discretizations are not identical nested classes; ratios need not increase monotonically. Agreement is a numerical sensitivity check, not a continuous-density guarantee.

C. Additional Derivations

C.1. An identifiability lower bound

Proposition 1. *Let two allowed parameters induce $N(\mu_0, I)$ and $N(\mu_1, I)$, with targets t_0, t_1 . Any interval covering each target with probability at least $1 - \alpha$ has expected length under the first law at least*

$$|t_1 - t_0|[1 - 2\alpha - \text{TV}(P_0, P_1)]_+, \quad (19)$$

where $\text{TV}(P_0, P_1) = 2\Phi(\|\mu_1 - \mu_0\|/2) - 1$.

Proof. Under P_0 , the probability of containing t_1 is at least $1 - \alpha - \text{TV}(P_0, P_1)$ by the definition of total variation. The union bound gives simultaneous containment of both targets with probability at least the displayed positive part. The length on that event is at least their separation. The total variation formula follows by projecting onto the line between the Gaussian means. $\square$

This standard two-point argument also applies to different allowed nuisance settings. Exact null directions give a nonvanishing uncertainty floor even with perfect image

prediction. It does not assert that a finite numerical search finds the largest possible separation.

Corollary 1 (Fixed-length Gaussian testing bound). *Under $y \sim N(\mathcal{A}\rho, I)$ and $\|\rho - \rho_0\| \leq B$, any interval of deterministic length $2q$ covering every target $\langle \ell, \rho \rangle$ with probability at least $1 - \alpha$, $0 < \alpha < 1/2$, satisfies, for each nonzero $h \in H$,*

$$q \geq |\langle \ell, h \rangle| \min \left\{ \frac{B}{\|h\|}, \frac{z_{1-\alpha}}{\|\mathcal{A}h\|} \right\}, \quad (20)$$

where the second ratio is infinite when $\mathcal{A}h = 0$.

Proof. Take $\rho_{\pm} = \rho_0 \pm \lambda h$ with $\lambda \|h\| \leq B$ and $\lambda \|\mathcal{A}h\| < z_{1-\alpha}$. The observation laws have total variation less than $1 - 2\alpha$. If $2q < 2\lambda |\langle \ell, h \rangle|$, the two target-containment events are disjoint, whereas their respective coverage probabilities force total variation at least $1 - 2\alpha$, a contradiction. Taking the limit at the strict boundary proves the result. $\square$

This is a classical testing argument, not a new minimax theorem. We evaluate $h = \ell - \mathcal{A}^*w$ and $h = \ell$ and retain the larger lower bound. Analytic target integrals and quadrature error bounds evaluate the pairings and norm denominators. The latter use upper bounds, preserving conservatism in real arithmetic; roundoff remains unvalidated. The result concerns deterministic interval length, unlike the preceding expected-length statement. Nominal poses form a submodel of the bounded-pose class, but this lower bound alone cannot establish sharpness for the larger class.

The same argument gives a generic efficiency certificate, following the optimal-recovery connection (Donoho, 1994; Armstrong & Kolesár, 2018). Let $U = \mathcal{F}_H(w)$ and $L > 0$ be a feasible fixed-pose dual value, with $(U - L)/U \leq \varepsilon$. Scaling its dual vector by $\kappa = z_{1-\alpha}/z_{1-\alpha/2}$ makes it feasible for the testing bound. Thus the optimum deterministic half-length q^* obeys $q^* \geq \kappa L$, whereas the reported half-width satisfies $q \leq U$. Consequently

$$\frac{q}{q^*} \leq \frac{z_{1-\alpha/2}}{z_{1-\alpha}(1 - \varepsilon)}. \quad (21)$$

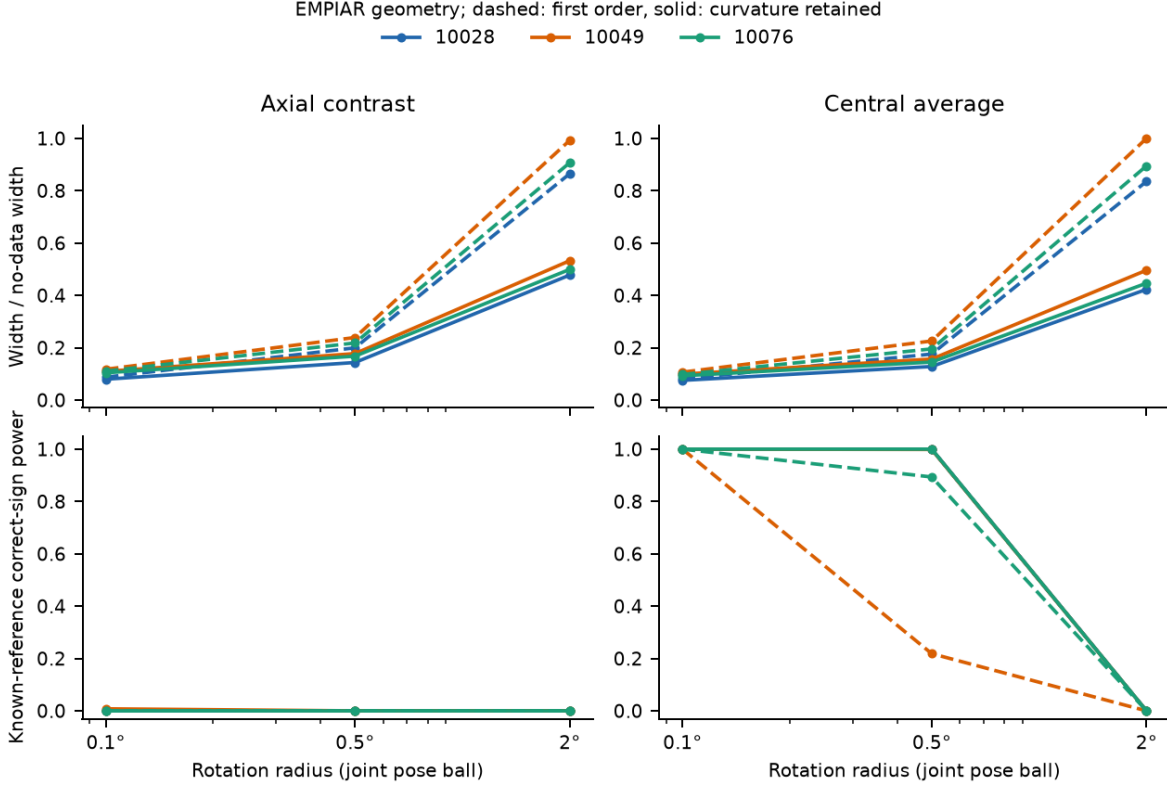


Figure 2. Curvature retained versus a first-order expansion with bounded second-order remainder, under the same density and pose classes. Dashed curves use the first-order procedure; solid curves retain curvature. The lower panels use the independent deposited-map generator under a coherent allowed rotation. Sign power and interval width expose limitations that high coverage alone conceals. These intervals precede the shared-density post-audit.

For $\alpha = 0.05$ and $\varepsilon = 0.005$, this is 1.198. The comparison applies to the continuous fixed-pose class; a post-hoc pose enlargement has no such gap certificate.

C.2. Gaussian density-energy Gram matrix

For isotropic paired features, let

$$P_{jk} = \left(\frac{2\pi\sigma_j^2\sigma_k^2}{\sigma_j^2 + \sigma_k^2} \right)^{3/2} \exp \left[-\frac{\|\mu_j - \mu_k\|^2}{2(\sigma_j^2 + \sigma_k^2)} \right], \quad (22)$$

$$Q_{jk} = P_{jk} \text{ with } \mu_j - \mu_k \text{ replaced by } \mu_j + \mu_k.$$

Integrating products of Gaussians and applying Parseval gives real and imaginary coefficient blocks $2(P + Q)$ and $2(P - Q)$, with zero cross block. A zero-centered imaginary coefficient is exactly null. Numerically excluded eigenmodes are reported; exclusion changes the stated Gaussian subspace. In an ambient audit they remain available to the larger space rather than being silently assigned zero uncertainty.

C.3. Uniform nonlinear remainder

For a Gaussian of width σ at radial distance r , bounds on value, gradient and Hessian are

$$g_0(r) = e^{-r^2/(2\sigma^2)}, \quad (23)$$

$$g_1(r) = rg_0(r)/\sigma^2, \quad (24)$$

$$g_2(r) = (1 + r^2/\sigma^2)g_0(r)/\sigma^2. \quad (25)$$

On an interval of possible distances, g_0 is largest at the lower endpoint; g_1 and g_2 attain their largest value at the point nearest σ . A rotation radius $a \leq \pi$ gives displacement at most $2\|k\|\sin(a/2)$, speed at most $a\|k\|$ and acceleration at most $a^2\|k\|$. A shift radius s gives phase speed at most $2\pi\|q\|s/\text{box}$.

The product rule bounds the second derivative by the sum of squared-speed times g_2 , acceleration times g_1 , twice speed times phase-speed times g_1 , and squared phase-speed times g_0 . Sum conjugate-pair bounds and apply the fixed CTF/noise scale. A coefficient-column Frobenius bound controls the operator term; the pilot-weighted bound controls its pilot term. Taylor’s integral formula gives Equation (14). Evaluating a Hessian only at the nominal pose

would not establish this uniform result.

C.4. Retaining quadratic pose terms

The isotropic quadratic remainder can dominate an interval. A development extension retains the curvature and uses a cubic remainder instead. Put $E_{iab} = \partial^2 A_i / \partial u_{ia} \partial u_{ib}$ at zero, $Q_i(w)_{ab} = w_i^\top E_{iab} \rho_0$, and let $U_i(w)$ collect columns $E_{iab}^\top w_i$ over all ordered pairs. Since $\|u_i \otimes u_i\|_F \leq 1$, a valid bias bound is

$$\begin{aligned} b_2(w) = & B \|\ell - \mathcal{A}^* w\| + \sum_i \|J_i^\top w_i\| \\ & + B \sum_i \|T_i(w)\|_F + \frac{1}{2} \sum_i \|Q_i(w)\|_F \\ & + \frac{B}{2} \sum_i \|U_i(w)\|_F + \sum_i \gamma_{3i} \|w_i\|. \end{aligned} \quad (26)$$

For a Fourier voxel column, define $T(q, x) = 2\pi a^3 \|k\| \|x\| / d$. Its third derivative along any allowed pose segment is bounded by $|C|[L^3 + 3LH + T]$ before whitening. The three summands follow by differentiating the phase exponential three times. Pilot contraction and a column Frobenius bound give H_{3i} and K_{3i} , hence $\gamma_{3i} = (H_{3i} + BK_{3i})/6$. Taylor’s integral remainder and Cauchy–Schwarz then give Theorem 1 with b_2 . Sharing of density errors and rank-one quadratic pose constraints are still relaxed.

Analytic column Hessians and pixel-space Gram factors implement these Euclidean groups. Mixed finite differences and independent nonlinear projections test the implementation. The full three-geometry, two-target comparison is shown in Figure 2. The spectral post-audit follows Equation (18); its proof is the operator-norm bound applied to the concatenated vectors $[\sqrt{d_j} v_j]_j$, whose squared norm is at most $\sum_j d_j$.

C.5. An independent noise-scale upper bound

Known noise is not necessary if an independently calibrated upper bound is available. A standard construction assumes independent calibration coordinates $z_j = \mu_j + \sigma e_j$, $e_j \sim N(0, 1)$, with arbitrary fixed means and independent inference noise. For ν coordinates, set

$$\bar{\sigma} = \|z\| / \sqrt{\chi_{\nu, \beta}^2}. \quad (27)$$

The probability that $\bar{\sigma} \geq \sigma$ is at least $1 - \beta$: the central chi-square pivot is exact at zero mean, and a noncentral chi-square has a smaller lower tail. Conditional on calibration, weights may use $\bar{\sigma}$. On this event their inference standard deviation is at most $s_+ = \bar{\sigma} \|w\|$. For $\alpha < 1/2$, $q(s_+, b, \alpha) \geq b$, and Gaussian coverage decreases with the actual noise standard deviation up to s_+ .

Integrating the conditional bound yields coverage at least $(1 - \alpha)(1 - \beta) \geq 1 - \alpha - \beta$.

This extension does not justify independence of windowed Fourier pixels or estimate arbitrary covariance. With 128 calibration coordinates and $\beta = 0.01$, a 20,000-draw development control gives scale-bound failure 0.0103 under independence. Equicorrelation 0.5 raises that failure to 0.503. Such a calibration set violates the premise even if a particular resulting interval remains conservative. Experimental whitening remains unresolved.

C.6. Simultaneous targets and sensitivity budgets

For K targets and parameter-budget choices fixed independently of inference noise, applying Theorem 1 with α/K gives simultaneous coverage by the union bound. Correlated Gaussian maxima can reduce the noise component if the joint weight covariance is accounted for. Selecting the most favorable point on a sensitivity surface after inspecting inference data requires such simultaneous control. A density radius supplied by simulation truth is an oracle control and cannot be reported as a practical calibration method.

D. Survey and Experimental Audit Trail

The repository’s living survey distinguishes density uncertainty, pose uncertainty, conformational variability/populations, map-processing confidence and atomic-model uncertainty. Its broad Europe PMC retrieval ledger contains 16,969 deduplicated records, mostly application papers; that count is not a count of reviewed methods. The curated list contains 89 candidate works at this draft checkpoint, with local full-text availability recorded separately. Recent items include LocScale-2.0, CryoDiff, CryoSplat, GEM, measurement-limited heterogeneity, population-counting work and the CAHRA dataset-construction preprint. Preprints and final publications are distinguished, and titles/versions are checked against primary sources.

The full experimental protocol draft records unimplemented comparisons and required failure controls. Fixed-pose resampling must be distinguished from rerunning alignment; a mathematical diagonal-Gaussian control must be distinguished from reproducing Ullrich et al.’s full software; cryoDRGN backprojection must be distinguished from training a neural model. Exposure-group splits do not undo the dependence introduced by full-data consensus poses. All development results, including the initially incomplete covariance-factor benchmark, remain identifiable by filename and status.

E. Continuous Fourier Audit

The sinc identity follows by integrating the two conjugate expansions of a real Fourier field. For a target component with center μ and width σ_ℓ , the one-dimensional truncated transform is

$$e^{-2\pi i k \mu - 2\pi^2 \sigma_\ell^2 k^2} \times [\Phi((1/2 - \mu)/\sigma_\ell + 2\pi i \sigma_\ell k) - \Phi((-1/2 - \mu)/\sigma_\ell + 2\pi i \sigma_\ell k)]. \quad (28)$$

Products over coordinates and sums over target components give $\mathcal{A}\ell$. Gaussian-product identities give $\|\ell\|^2$. Independent tensor quadrature verifies these analytic expressions, while an independent conic problem formed from the joint Hilbert Gram of $[\ell, \mathcal{A}_1^*, \dots, \mathcal{A}_m^*]$ checks optimization and dual feasibility.

For r Gauss–Legendre nodes on a unit interval, the classical error constant is $C_r = (r!)^4 / [(2r+1)((2r)!)^3]$ (NIST, 2026). Put $M_a = \max_j |k_{ja}|$ and $E_r = \sum_a C_r (4\pi M_a)^{2r}$. Positive tensor weights ω_t satisfy

$$\left| \|f_w\|^2 - \sum_t \omega_t f_w(x_t)^2 \right| \leq E_r \left(\sum_j |c_j| \right)^2. \quad (29)$$

To verify the bound, expand f_w^2 into exponentials with coordinate frequencies at most $2M_a$. Apply the one-dimensional derivative remainder and telescope the three integration/quadrature operators. Their positive weights have total mass one, and the expansion’s coefficient absolute sum is bounded by $(\sum_j |c_j|)^2$.

Similarly, the error in the continuous Gram action obeys

$$\|(G - G_Q)w\| \leq \sqrt{2} E_r \|C/\sigma\|_2 \sum_j |c_j|. \quad (30)$$

Adding the scalar remainder to the squared primal residual and the vector remainder to the dual feasibility denominator preserves the continuous objective bracket in real arithmetic. Two type-3 NUFFTs apply G_Q . At frequency radius five, quadrature order 40 makes the analytic kernel remainder at most 3.21×10^{-23} before coefficient scaling. This quadrature theorem does not certify FINUFFT’s floating-point error; requested tolerance 10^{-12} is checked numerically against independent sinc Grams.

An orthonormal constant-cell coefficient is scaled by the square root of inverse cell volume. Its exact transform includes $\prod_a \text{sinc}(k_a/d)$ and a half-cell phase relative to the centered FFT grid. This prevents a point-sample approximation from being mislabeled exact continuous physics. For an orthogonal cell projection P , $\|h\|^2 = \|Ph\|^2 + \|(I - P)h\|^2$. The omitted component is the continuous analogue of the omitted-dictionary term, and can affect coverage even when successive grid widths look similar.

E.1. Continuous nonlinear-pose bound

Proposition 2. Let $H = L^2([-1/2, 1/2]^3)$, $\|\rho - \rho_0\| \leq B$, $P = \|\rho_0\|$ and $\|u_i\| \leq 1$. Put $h = A_0^* w - \ell$. Suppose L_{12} bounds the continuous norm of the sum of first and half-scaled second pose-adjoint fields. For rotation radius a and shift radius s_t in field units, define $R = \sqrt{3}/2$ and

$$L_{iq} = 2\pi(a \|k_{iq}\| R + s_t \|q\|), \quad (31)$$

$$H_{iq} = 2\pi a^2 \|k_{iq}\| R, \quad T_{iq} = 2\pi a^3 \|k_{iq}\| R, \quad (32)$$

$$K_{3i}^2 = \sum_q |C_{iq}/\sigma|^2 (L_{iq}^3 + 3L_{iq}H_{iq} + T_{iq})^2. \quad (33)$$

Then Equation (11) bounds the scalar bias for every density and pose in the class. Combined with Theorem 1, it yields a continuous-density interval under the same independent Gaussian-noise assumptions.

Proof. The exact bias splits as $\langle h, \delta \rangle + \langle (A(u)^* - A_0^*)w, \rho_0 + \delta \rangle$. The first term is at most $B \|h\|$. The second pairs a Taylor polynomial plus remainder with a density of norm at most $P + B$. Along a rotation segment, the first three frequency-derivative norms are bounded by $a \|k\|$, $a^2 \|k\|$ and $a^3 \|k\|$; the shift phase is linear. The third exponential derivative is bounded by $L^3 + 3LH + T$. The cube has unit volume, so summing squared column bounds gives the realified Hilbert–Schmidt bound K_{3i} . Taylor’s integral remainder contributes $1/6$; Cauchy–Schwarz with w_i completes the bound. $\square$

To compute L_{12} , use orthonormal symmetric vectorization of the Hessian: diagonal entries are unchanged and off-diagonals are multiplied by $\sqrt{2}$. Its coefficient vector has norm $\|u_i\|^2 \leq 1$, so fifteen columns replace twenty-five ordered columns exactly. The derivative fields are polynomials of degree at most two times Fourier exponentials. Their Gram products have coordinate degrees at most four.

For quadrature order r , $n = 2r$, and $\omega_a = 4\pi \max_j |k_{ja}|$, define

$$D_\beta(\omega) = \sum_{l=0}^{\min(\beta, n)} \binom{n}{l} \frac{\beta!}{(\beta-l)!} 2^{l-\beta} \omega^{n-l}, \quad (34)$$

$$E_{\text{poly}} = C_r \sum_a \max_{0 \leq \beta \leq 4} D_\beta(\omega_a). \quad (35)$$

Leibniz differentiation, the positive Gauss remainder kernel and tensor telescoping bound each monomial-exponential integration error by E_{poly} . If column j has absolute polynomial-Fourier coefficient sum at most M_j , the Gram-entry error is at most $E_{\text{poly}} M_j M_l$ and its operator norm at most $E_{\text{poly}} \sum_j M_j^2$. After dividing each column and its M_j by $\sqrt{d_{\text{block}(j)}}$, add that error to the largest quadrature-Gram eigenvalue in Equation (18). Separate and joint spectral bounds, and a padded Frobenius triangle bound, each remain valid; their minimum supplies L_{12} .

This integrates known adjoint fields, not the unknown density. Direct nonlinear Fourier fields check derivatives and cubic remainders independently. Exact constant-cell projections test reference generators. At a fixed nonzero pose, sinc integrals give the feasible worst bias over the entire continuous density ball, $|\langle \rho_0, (A(u)^* - A_0^*)w \rangle| + B \|A(u)^*w - \ell\|$; it is a lower bound on the global pose supremum. Floating-point and NUFFT tolerances remain numerical checks rather than validated arithmetic.

E.2. Integrated-moment cubic remainder

The earlier K_{3i} uses the maximum spatial radius everywhere. A sharper scalar-estimator bound integrates known Fourier columns over the cube. Put

$$c_2 = 12^{-1/2}, \quad c_4 = (3/12^2)^{1/4}, \quad c_6 = (15/12^3)^{1/6}. \quad (36)$$

For any $v \in \mathbb{R}^3$, $\|v^\top x\|_{L^p(D)} \leq c_p \|v\|$ for $p = 2, 4, 6$. To see this, expand the even moment under unit-volume cube integration. Odd-coordinate terms vanish, and each remaining term is nonnegative. The coordinate moments $1/12$, $1/80$ and $1/448$ are bounded by those of a centered Gaussian with variance $1/12$. Termwise comparison gives the stated constants. This is an integration identity for the operator, not a Gaussian model of the density.

Proposition 3 (Integrated scalar remainder). *Under Proposition 2’s density and pose classes, define*

$$L_p(k, q) = 2\pi(a \|k\| c_p + s_t \|q\|), \quad (37)$$

$$H_p(k) = 2\pi a^2 \|k\| c_p, \quad T_2(k) = 2\pi a^3 \|k\| c_2, \quad (38)$$

$$D(k, q) = L_6(k, q)^3 + 3L_4(k, q)H_4(k) + T_2(k). \quad (39)$$

For $c_{iq} = (w_{R,iq} + iw_{I,iq})C_{iq}/\sigma$, the uniform scalar cubic remainder is bounded by

$$r_{\text{mom}}(w) = \frac{B+P}{6} \sum_{i,q} |c_{iq}| D(k_{iq}, q). \quad (40)$$

It can replace the cubic term in Equation (11), preserving the same conditional coverage guarantee.

Proof. Along a normalized pose segment, the phase $\phi(t, x)$ has derivative norms $\|\phi'\|_p \leq L_p$, $\|\phi''\|_p \leq H_p$ and $\|\phi'''\|_2 \leq T_2$. This follows from $\|k^{(j)}(t)\| \leq a^j \|k\|$ and linear detector translation. The exact third derivative of $e^{i\phi}$ is $e^{i\phi}(i\phi''' - 3\phi'\phi'' - i(\phi')^3)$. Minkowski and Hölder give its L^2 norm at most D . Triangle inequality with the fixed coefficients c_{iq} , Taylor’s integral weight $1/6$, and Cauchy–Schwarz against a density of norm at most $B+P$ prove Equation (40). Moreover, $\sum_q |c_{iq}| D_{iq} \leq \|w_i\| (\sum_q |C_{iq}/\sigma|^2 D_{iq}^2)^{1/2}$, so retaining actual weights is no worse than this integrated Hilbert–Schmidt bound. $\square$

The refinement applies classical inequalities to the Fourier-slice operator. Independent numerical integration checks the moment envelope, matrix-exponential derivatives, and exact nonlinear remainder. Those tests complement the proof; they are not exhaustive verification or validated arithmetic. The first development evaluation uses all original weights without refitting. A subsequent frozen protocol reserves different acquisition subsets and new signal/noise seeds for the one- and two-degree settings.

F. Uncertainty Targets in the Recent Literature

The following distinctions guide baseline selection. They do not rank methods that solve different problems on one inappropriate score. The accompanying dated survey has 93 curated candidates across reconstruction, poses, heterogeneity, map validation, atomic ensembles and general inverse inference; its source manifest distinguishes retrieved material from targeted critical reading and inaccessible full text. This is a scoped critical survey rather than a claim to have exhaustively reviewed all cryo-EM applications. Versioned preprints are identified separately from published articles.

Additional local and atomic validation. The recent Gold-standard local toolbox compares half-map phase and amplitude, with explicit independent-noise assumptions in its SNR derivation (Vargas et al., 2026). EMMIVox combines molecular force fields, map likelihoods and half-map-informed uncertainty for atomic refinement (Hoff et al., 2024). Neither output is the fixed raw-particle density functional studied here.

Direct statistical uncertainty prior art. The probabilistic Fourier-slice model of Ullrich et al. (2020) is a close reconstruction-uncertainty comparator and explicitly recognizes model bias. Joint likelihood soft modes (Rangan et al., 2024) expose pose–density ambiguities. Lai et al. (2023) connect empirical Bayes and sequential MCMC filtering to dynamic image reconstruction, including scalar confidence sets and hybrid resampling. Their approximate fitted-family resampling and posterior computations differ from the present uniform conditional bounded-class statement; they cannot be dismissed as deterministic reconstruction methods. Neither general bias-aware inference nor the distinction between posterior variance and model error originates here.

Pose inference versus pose confidence. Posterior-averaged orientation estimation (Xu et al., 2026) and probabilistic orientation search in SIMPLE (Van et al., 2025) improve pose estimation through different statistical or optimization constructions. Neither an orientation distribution nor a stochastic search neighborhood should be relabeled a

calibrated per-particle confidence ball without verifying its guarantee. Our prescribed local balls avoid that claim, but consequently do not solve practical pose calibration.

Cryo-forum uses quaternion-distribution dispersion for error ranking and filtering (Chung, 2023). ARCHER learns reference-conditioned pose scores and evaluates cross-specimen transfer and junk rejection (Nguyen & Pham, 2026). CESPED standardizes experimental poses obtained by refinement, and explicitly acknowledges uncertainty in those labels (Sanchez-Garcia et al., 2024). CryoPARES transfers alignments between related experiments (Sanchez-Garcia et al., 2026); CryoFastAR combines multiview pose prediction, synthetic training and experimental fine-tuning (Zhang et al., 2025). These approaches address efficiency and different transfer settings. Their quality scores and angular errors cannot supply a uniform maximum-error radius for every particle in a new stack. The 0.1° and 0.5° budgets in our frozen study are sensitivity settings, not experimentally established alignment accuracy.

Physical heterogeneity versus estimation error. RECOVAR estimates covariance and uses kernel regression to account for noisy latent coordinates (Gilles & Singer, 2025). DynaMight and 3DFlex represent motion (Schwab et al., 2024; Punjani & Fleet, 2023). Particle-counting work explicitly studies when assignments misestimate state populations (Evans et al., 2026). These motivate separate tests of population recovery, structural variation and uncertainty in a homogeneous density. The present bounded-density target covers only the last. CryoDRGN-AI addresses joint ab initio reconstruction (Levy et al., 2025); fixed-pose homogeneous training does not reproduce that task.

DynaMight’s held-out deformation disagreement and independent-half training assess motion errors; 3DFlex separates lower-frequency motion training from higher-frequency reconstruction validation. Their protocols must be examined at the level of shared components and evaluated frequencies. Mattingly et al. (2026) study how the image model limits learnable conformational populations, using information-based selection of ensemble granularity. That prior-averaged population target differs from our fixed homogeneous density functional. CryoBench likewise distinguishes representation learning, conformation assignment and volume recovery with several reference-based FSC metrics (Jeon et al., 2025). We do not equate our three homogeneous acquisition populations with its heterogeneous benchmark tasks.

Map validation and learned priors. False-discovery confidence maps test density significance against a noise reference (Beckers et al., 2019). Directional resolution measures acquisition anisotropy (Vilas et al., 2020); independent-particle tests assess map fit (Ortiz et al., 2020). Bias/variance

analyses explain why validation requires more than internal agreement (Sorzano et al., 2022). Blush introduces learned reconstruction regularization (Kimanus et al., 2024), while LocScale-2.0 attaches confidence to enhancement relative to its stated proxy (Bharadwaj et al., 2026). Matched evaluations must retain these targets. A sharper-looking map or a significant positive voxel is not itself a bound on error relative to unknown density.

CryoRes predicts local resolution using deposited and algorithm-derived training labels (Dai et al., 2023). Such supervision inherits the meaning of those labels; it does not transform a resolution score into a confidence interval. Similarly, held-out image prediction measures agreement with new observations, and can remain favorable when two reconstructions share unsupported density directions. Ground-truth simulation, new-exposure prediction, directional diagnostics and independent structural evidence therefore complement one another.

Atomic ensembles and independent modalities. Experiment-guided AlphaFold3 conditions structural generation on reconstructed ESP maps and other measurements (Maddipatla et al., 2026). Its atomic-ensemble target differs from raw-particle density inference. The multimodal examples also illustrate why improved map agreement need not imply improved fit to a separate measurement modality. Independent evidence is relevant to assessing structural interpretations that the density measurements alone leave ambiguous.

Simultaneous inverse inference. Constrained test-inversion methods already address joint confidence regions for multiple functionals (Batlle et al., 2025). A single data-consistency region can support simultaneous projections, whereas separately calibrated marginal intervals generally require an appropriate correction. Our present scalar-target theorem is not a new general framework for simultaneous inference, and a colored map of marginal intervals must not be presented as simultaneous coverage.

Conformal calibration and its sampling unit. Posterior-variance calibration can provide error bounds even when a posterior sampler is approximate (Narnhofer et al., 2024); its theorem uses independent identically distributed signal-observation pairs and conditions on a variance bin. QUTCC learns quantile-dependent image predictions and calibrates marginal pixel errors from paired reference images (Ye et al., 2026). These methods can be useful without our fixed-signal radius assumption, but require a calibration population with the stated sampling relationship to deployment. Voxels or noise draws from one molecule are not automatically independent calibration molecules. Neither marginal population coverage nor variance-bin coverage is the same as coverage

Table 4. Joint ambient/pose interval width divided by the no-data width. Columns 2–4 use a 100-iteration cap. The last column repeats 2° with up to 500 iterations; EMPIAR-10049 retains a 0.548% gap, above the 0.5% tolerance. Others meet tolerance. These are different optimization budgets, not separate independent replications.

EMPIAR	0.1°	0.5°	2°	2° , longer
10028	0.087	0.200	0.866	0.865
10049	0.118	0.239	0.996	0.992
10076	0.110	0.218	0.910	0.907

uniformly over every density in a specified class.

Open questions suggested by the survey. Particularly consequential gaps are absolute pose-confidence calibration under reference error; uncertainty in conformational populations when candidate structures are imperfect; validation of detail introduced by learned priors using independent particles; and uncertainty propagation through CTF fitting, image preprocessing, alignment and atomic interpretation. Acquisition chosen for a specific structural feature is another possible application of adjoint sensitivity. These are directions inferred from the literature, not priority claims or completed capabilities of this implementation. The present continuous audit addresses representation exclusion under stated nuisance assumptions; solving those assumptions empirically remains distinct work.

G. Additional Development Comparisons

Table 5. Frozen additional-exposure prediction: 4,096 particles per dataset from 115, 71 and 185 source groups absent from development. Models and cohort were locked before image access. Lower normalized prediction MSE is better. Supplied consensus poses/CTFs remain conditional; this does not measure unknown density coverage.

	EMPIAR	Gaussian	Voxel	Neural
	10028	0.835319	0.841379	0.829369
	10049	0.931579	0.932359	0.930739
	10076	0.874799	0.879078	0.868657

EMPIAR	Comparator minus neural	NMSE difference	Adjusted percentile interval
10028	Gaussian	0.005950	[0.004995, 0.006932]
10028	Voxel	0.012011	[0.010695, 0.013356]
10049	Gaussian	0.000840	[0.000659, 0.001023]
10049	Voxel	0.001620	[0.001366, 0.001875]
10076	Gaussian	0.006143	[0.005686, 0.006593]
10076	Voxel	0.010421	[0.009792, 0.011038]

All six prespecified paired differences are shown. Intervals use 10,000 whole-exposure bootstrap resamples and a Bonferroni adjustment for six contrasts (99.1667% marginal percentile intervals). This is a bootstrap approximation, not a finite-sample coverage theorem. No model selection uses these outcomes.

Table 6. Frozen continuous-density and pose validation: relative half-width ranges over both targets and all four geometry replicates per stack. All 96 settings and all in-class signals are retained. The density, noise and pose limits are supplied assumptions.

EMPIAR	Scale 0.03		Scale 0.07	
	0.1°	0.5°	0.1°	0.5°
10028	0.76–0.77	0.98–1.00	0.11–0.14	0.31–0.38
10049	0.77–0.78	0.93–0.95	0.14–0.17	0.38–0.46
10076	0.76–0.77	0.89–0.89	0.12–0.14	0.27–0.31

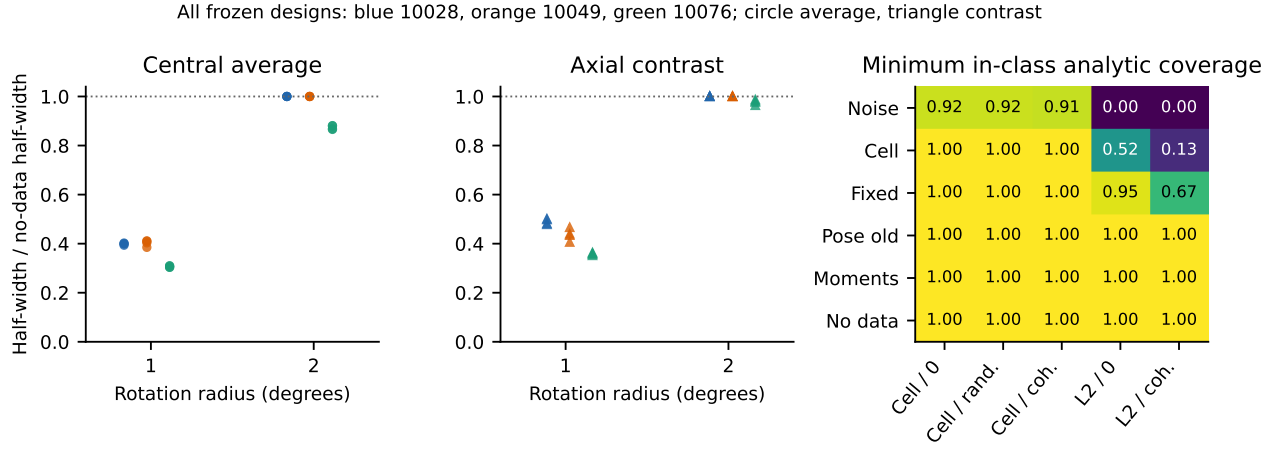


Figure 3. All 48 frozen integrated-moment settings on disjoint particle subsets with new simulation seeds. The original spatial-maximum pose correction falls back to no data throughout; integrated moments retain informative widths in 32 settings. Conservative coverage and poor cell-generator sign power remain limitations. “L2” denotes constructed continuous-density boundaries.

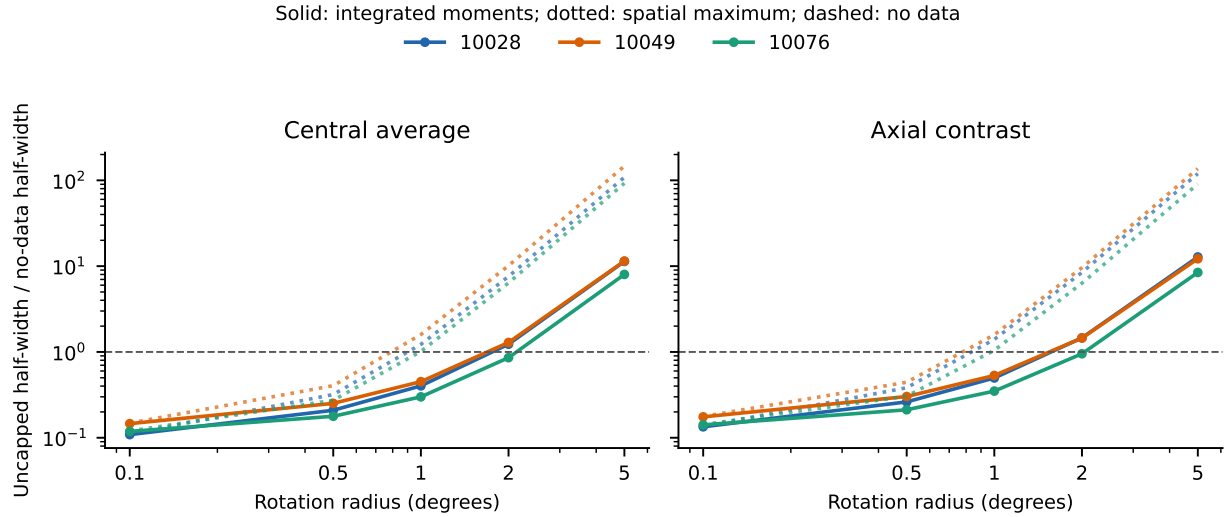


Figure 4. Post-hoc integrated-moment refinement for both broad targets and every tested pose radius. Curves show uncapped widths to expose conservatism; the implemented decision uses the no-data estimator above the dashed line. The solid curves use integrated Fourier-column moments and actual estimator weights; dotted curves use the older spatial-maximum cubic remainder. No weights are refitted.

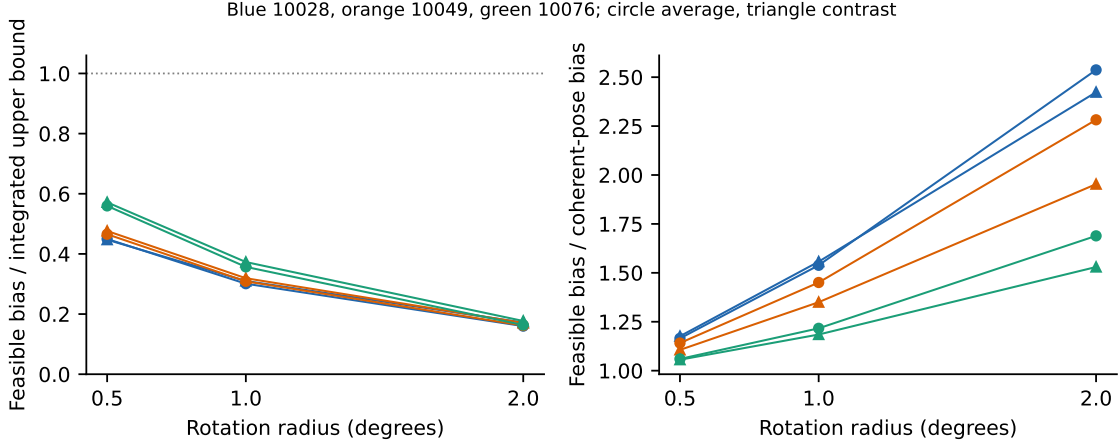


Figure 5. All 18 continuous nonlinear stress settings, retaining the strongest of six feasible searches per setting. The left panel compares numerical feasible bias with the integrated-moment upper bound; the right compares it with the earlier coherent-pose control. Surrogate optimization proposes poses only: final candidates eliminate the full continuous density ball using analytic sinc integrals. The largest independent direct-field relative discrepancy was 5.35×10^{-6} . Remaining upper/lower gaps and local-search limitations preclude a sharpness claim.

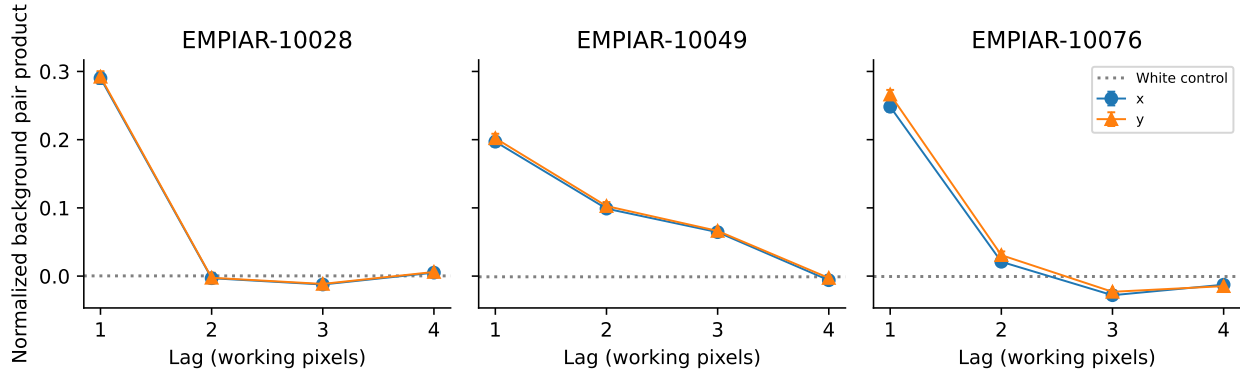


Figure 6. Descriptive background dependence in every additional-exposure image, after Fourier cropping of the downloaded particles to the working grid and before reconstruction windowing. We center and scale each image using pixels outside radius 0.43 of the box, then average products of background pixel pairs separated by each lag. Error bars are marginal 95% whole-exposure bootstrap intervals. Dotted lines show matched-statistic white-Gaussian controls. Background contamination and preprocessing prevent interpreting these values as pure-noise correlations or a confidence calibration.

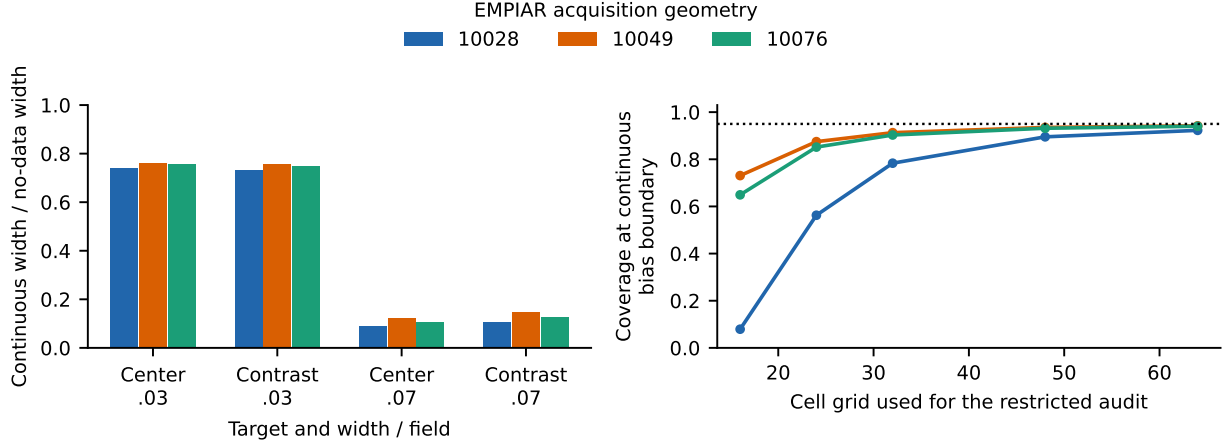


Figure 7. Left: optimized continuous fixed-pose widths for the full cube class. Right: cell-projected audit coverage at a constructed continuous bias boundary, for the broad central target and cell-fitted estimator. Reference-map coverage is one in these cases. The two panels answer distinct questions: useful width and omitted continuous directions.

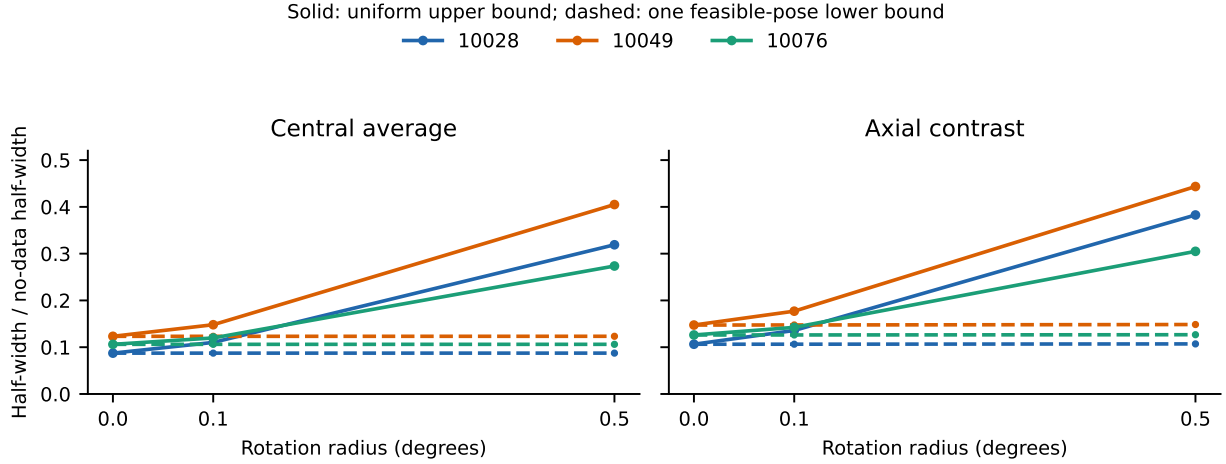


Figure 8. Continuous-density and nonlinear-pose development audit at target width 0.07 of the field. Solid curves are uniform conditional upper bounds for the same fixed-pose-optimized weights. Dashed curves use exact density-ball elimination at one allowed coherent rotation, providing feasible lower bounds on the required width, not global pose optima. Zero is the separate exact-pose control; the nonzero budgets also allow shifts of $0.01/24$ field units. The visible gap exposes conservatism.

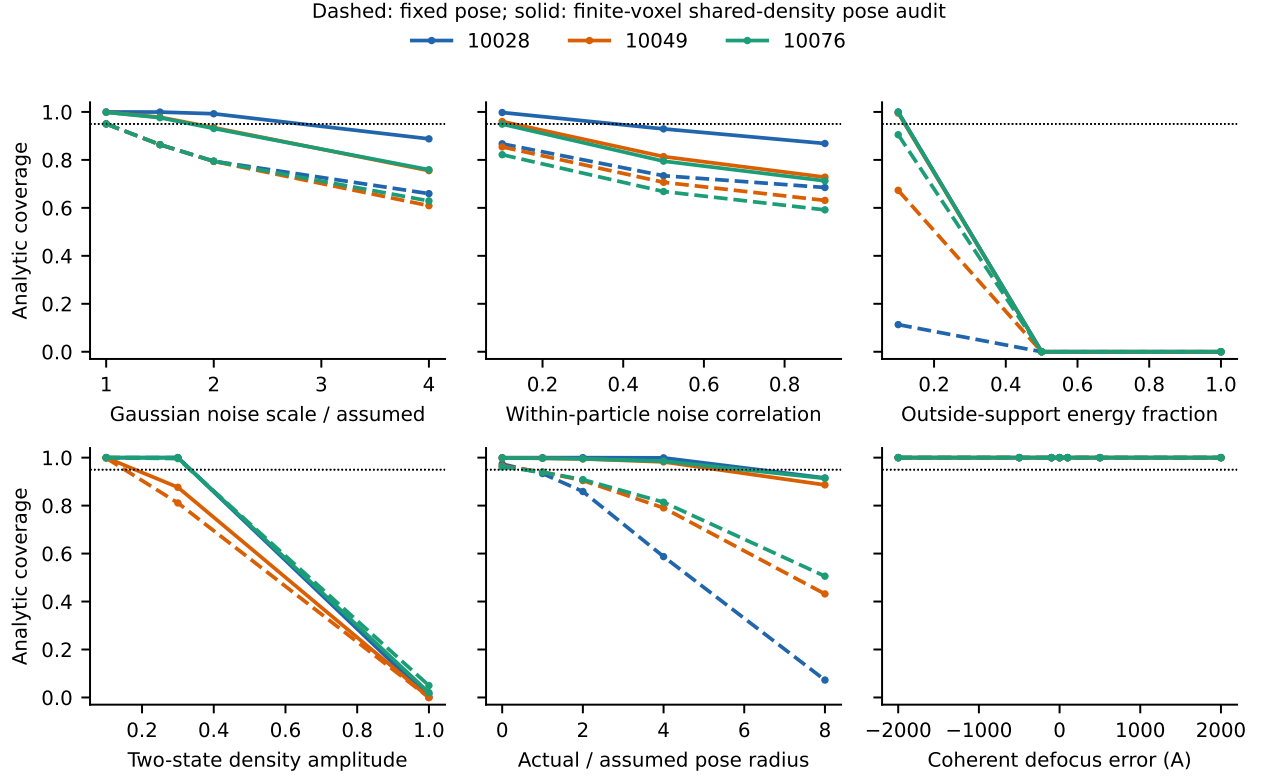


Figure 9. Deliberate assumption stresses for the finite-voxel central target. Curves share a signal within each geometry/control: the fixed-pose boundary for noise tests, a fixed-pose-estimator-directed support or mixture perturbation, or a saved nonlinear pose adversary. Dashed curves are fixed-pose certificates; solid curves include the shared-density pose audit. These are constructed diagnostic conditions, not experimental failure rates. All other controls remain in the result records.

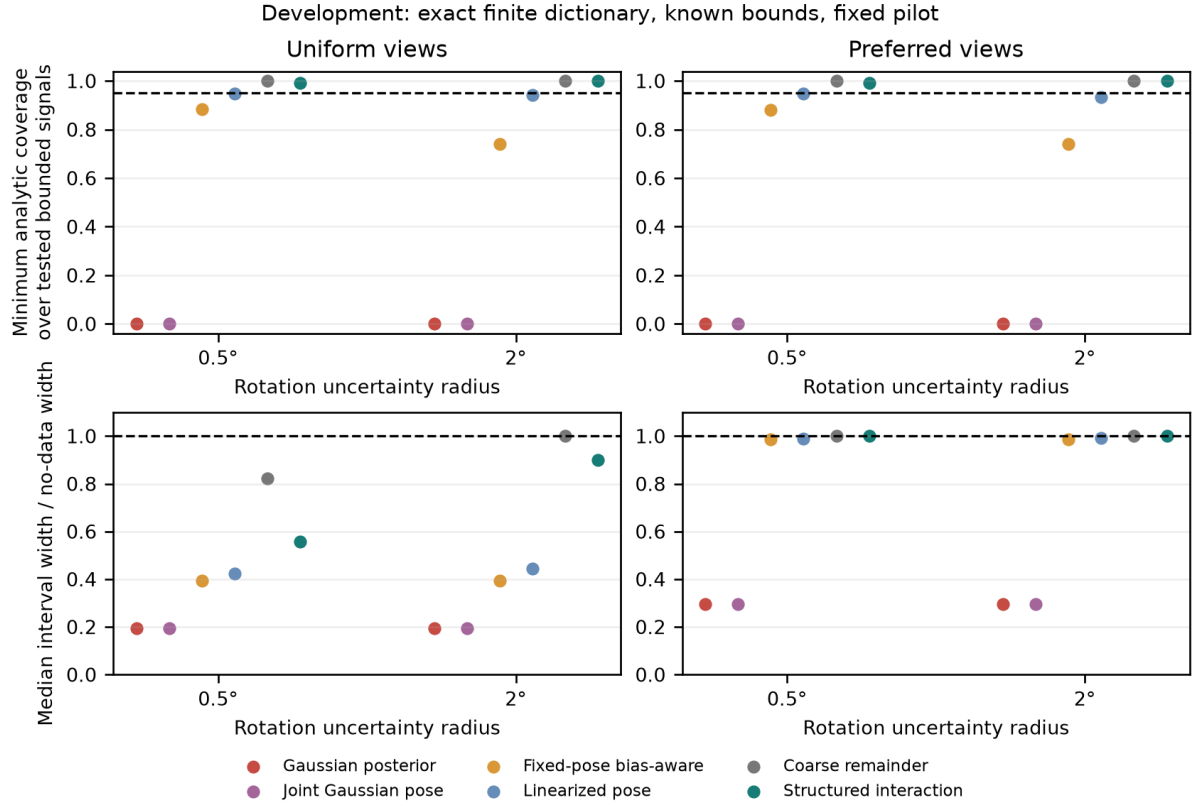


Figure 10. Exact-dictionary development tests with known bounds and an independent fixed pilot. Minimum coverage is over the finite tested set, including adversarial signals; it is not an empirical proof of the worst case. Widths count each independent design once. Gaussian posterior methods separately pass their correctly matched prior-predictive coverage check at 0.95. Their fixed-parameter boundary failures do not contradict Bayesian prior-average calibration.

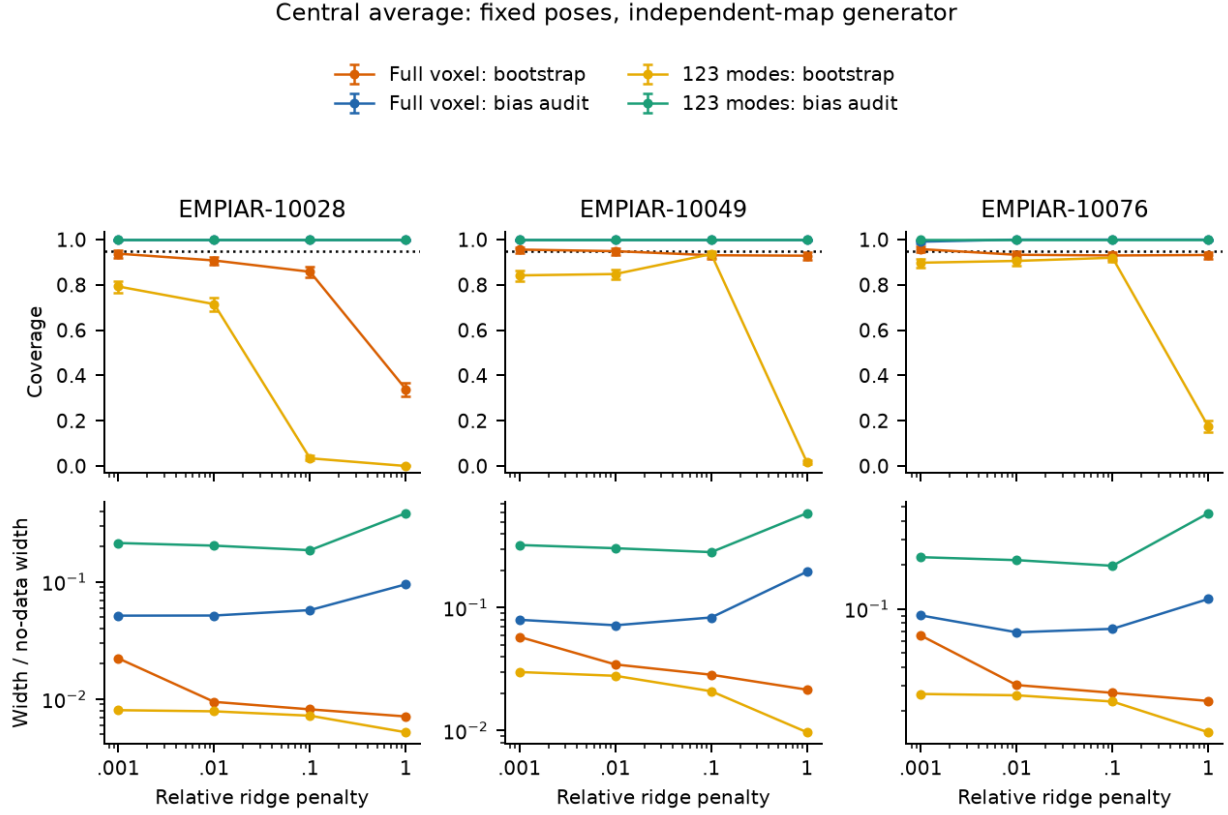


Figure 11. Central-average coverage and widths across all four ridge penalties. Each geometry uses its own fixed independent deposited-map generator. Error bars are binomial 95% intervals over independent measurement-noise draws, conditional on the reused group-bootstrap plan. Weak-regularization bootstrap can be narrow and near nominal; stronger regularization exposes bias. These tests do not establish empirical coverage for unknown experimental density.

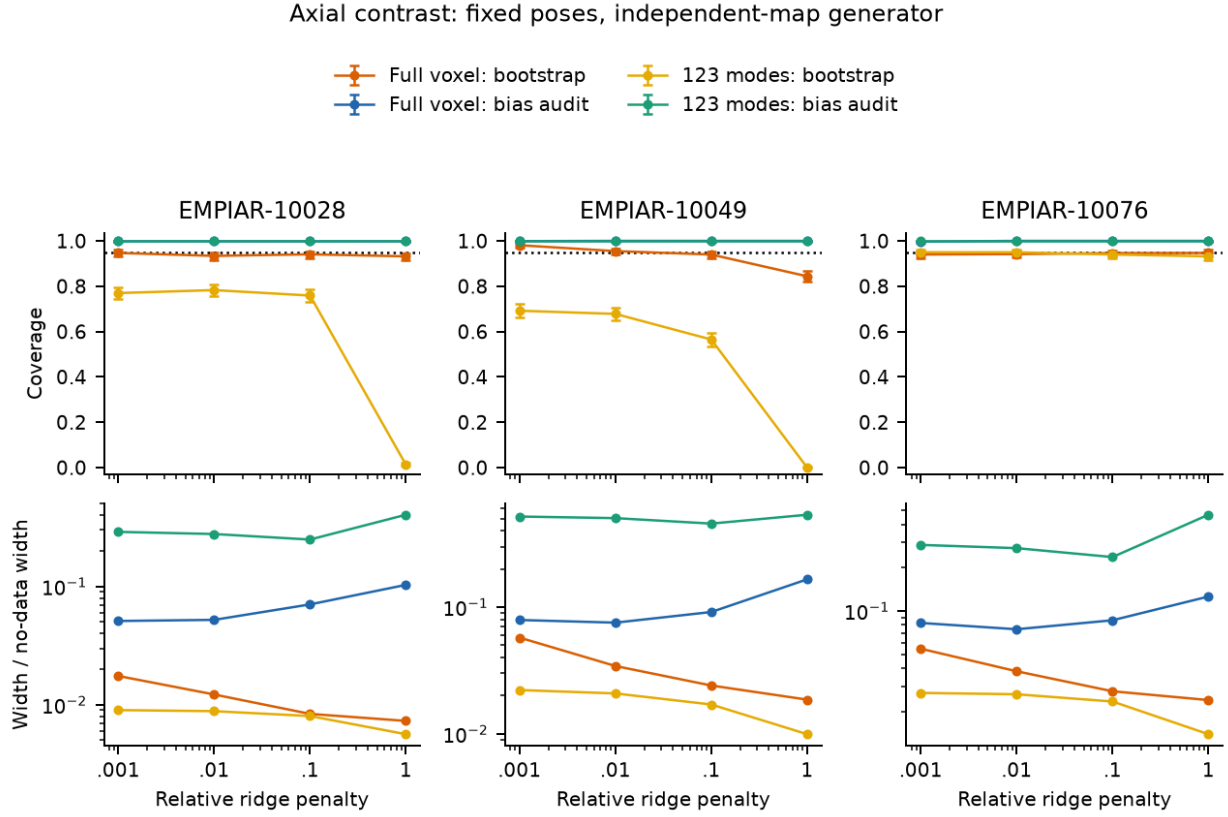


Figure 12. The same complete regularization comparison for the axial contrast. Both the estimand and the unknown simulation signal are fixed before noise replication. The full results also retain the smaller 33-mode dictionary, basic bootstrap and Gaussian covariance controls.

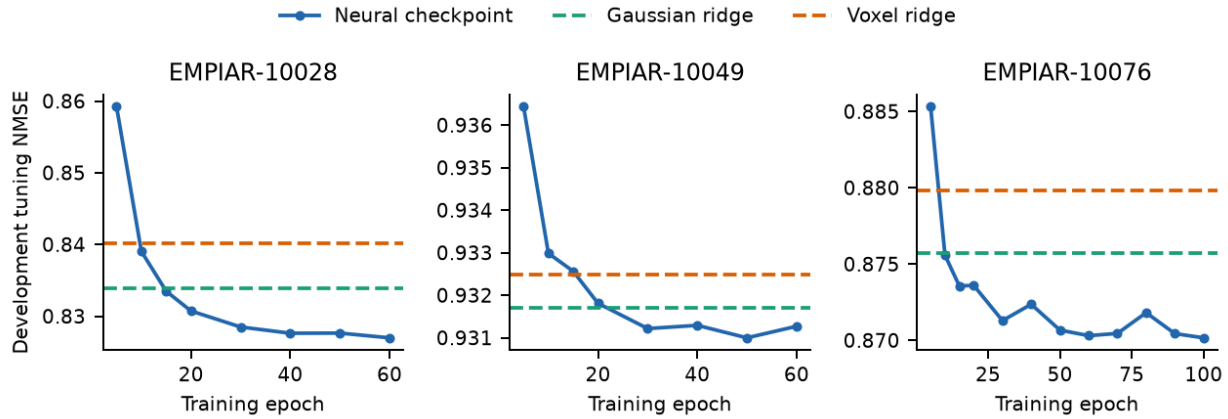


Figure 13. Development tuning error at all evaluated neural checkpoints, with the classical reconstruction controls. Checkpoint selection minimizes this tuning error; the test-group error is not used to select the checkpoint. Different training objectives and supplied consensus poses limit this comparison.